

CSE 440: Introduction to AI (Spring 2026)
Department of Computer Science and Engineering
Michigan State University

Instructor: Dr. Yu Kong

D2L: <https://d2l.msu.edu/d2l/home/2387354>

Piazza: <https://piazza.com/msu/spring2026/cse440> (Piazza is for all communications)

Location: Wells Hall B115

Time: Tuesday and Thursday, 10:20AM-11:40AM

Teaching Assistants:

Wenliang Guo, Bingheng Li, Karan Choudhary

Prerequisites: Prerequisite: (CSE 331) and (MTH 314 or ECE 280) **Restrictions:** Open to juniors or seniors in the College of Engineering or in the Computer Science Major or in the Lyman Briggs Computer Science Coordinate Major or in the Lyman Briggs Computer Science Major or in the Data Science Major. Please contact the instructor if you have any questions.

Textbook:

[1] Russell & Norvig, Artificial Intelligence: A Modern Approach. Pearson Education

Recommended Reading:

- Richard S. Sutton and Andrew G. Barto. Reinforcement Learning: An Introduction, second edition. The MIT Press
- Goodfellow et al., Deep Learning, MIT Press (<https://www.deeplearningbook.org>)
- Christopher M. Bishop. Pattern Recognition and Machine Learning (Information Science and Statistics). Springer
- Duda, et al., Pattern Classification, John Wiley & Sons

Description:

Fundamental issues in artificial intelligent systems. Data representation and mechanisms of reasoning. Search and constraint satisfaction. Agents and decision making. Application areas of AI and current topics. Students will learn basic concepts of artificial intelligence and get some hands on experience to solve real-life problems. Python will be used for all the assignments.

Objectives:

To equip students with a good knowledge of: (a) the search problems and algorithms in AI; (b) the decision-making process of AI; (c) data representation and reasoning on Bayesian networks; (d) machine/deep learning concepts. The concepts will be explained from a pattern recognition and computer vision perspective.

Course Topics:

- Markov Decision Process
- Reinforcement learning
- Bayesian network
- Machine Learning
- Deep learning

Grading:

The tentative weight associated with each grading component is as follows:

- Attendance – 10%
- Quiz – 10%
- Homework – 20%
- Lab – 25%
- Mid-term Exam – 15%
- Final Exam – 20%

Assignments and tests will be based on (a) topics covered during lectures, (b) PowerPoint files used in lectures, (c) contents of the textbook, and (d) reading material assigned by the instructor. Students are expected to take down notes during the lecture. **Students who miss 3 assignments or more will automatically Fail this course.**

Grading Policy:

- Assignments need to be turned in D2L before the due date. Late assignments for more than 24 hours will not be accepted.
- Make-up for exams will only be granted under exceptional circumstances. Instructor reserves the right to deny requests for make-up.
- No grade rounding in this class. For example, if a student receives 79.99, then we will NOT round it to 80.00.

Grade Assignment:

The grading scheme will be as follows:

- 90-100: 4.0
- 85-89: 3.5
- 80-84: 3.0
- 75-79: 2.5
- 70-74: 2.0
- 60-69: 1.5
- 50-59: 1.0
- 0-49: 0.0

Attendance:

- Attendance at all regularly scheduled class meetings is a requirement of this course. Students with unexcused absences for more than three consecutive class meetings prior to the middle of the term may be dropped from the course for non-attendance.
- We will randomly check attendance throughout the semester.
- Attendance cheating is certainly an incident of academic dishonesty and it could be subject to the filing of an academic dishonesty report (ADR).

Academic Integrity:

Article 2.III.B.2 of the Academic Freedom Report states that "The student shares with the faculty the responsibility for maintaining the integrity of scholarship, grades, and professional standards." In addition, the Department of Computer Science and Engineering adheres to the policies on academic honesty as specified in General Student Regulations 1.0, Protection of Scholarship and Grades; the all-University Policy on Integrity of Scholarship and Grades; and Ordinance 17.00, Examinations. (See [Spartan Life: Student Handbook and Resource Guide](#))

Therefore, unless authorized by your instructor, you are expected to complete all course assignments, including homework, projects, quizzes, tests and exams, without assistance from any source. You are expected to develop original work for this course; for example, you may not submit course work you completed for another course to satisfy the requirements for this course. Students who violate MSU academic integrity rules may receive a penalty grade, including a failing grade on the assignment or in the course. Contact your instructor if you are unsure about the appropriateness of your course work. (See also the [Academic Integrity](#) webpage.)

Examples of academic dishonesty include (but are not limited to):

- Copying another student's code, assignment solutions or quiz/exam answers
- Using code implemented by someone else intended to solve this class's assignments (i.e., don't get someone else to do your assignment for you)
- Using code independently implemented by someone else without attributing credit (i.e., you can use tools, libraries, or code snippets from the web, but only with proper citation and with permission of the instructor where appropriate)
- Using websites and sources, whose purpose is to provide assignment solutions
- Distributing course content without instructor permission
- Providing false information to the instructor about matters related to the course.
- Facilitating another student in any of these activities.

The Spartan Code of Honor Academic Pledge

As a Spartan, I will strive to uphold values of the highest ethical standard. I will practice honesty in my work, foster honesty in my peers, and take pride in knowing that honor in ownership is worth more than grades. I will carry these values beyond my time as a student at Michigan State University, continuing the endeavor to build personal integrity in all that I do.